Formula Sheet for Midterm Exam

On the exam I will provide the following equations (Note: The exam is a closed book exam and you are not allowed to bring this formula sheet on the exam.)

$$V_p = \sqrt{(V_0)^2 + (\frac{V_{+1} - V_{-1}}{2\omega_0 \Delta T})^2} \qquad \theta = \tan^{-1}(\frac{V_0}{\frac{V_{+1} - V_{-1}}{2\omega_0 \Delta T}})$$

$$v(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)$$

$$a_n = \frac{2}{T}\int_{t_0}^{t_0+T} v(t)\cos(n\omega_0 t)dt \qquad n = 0,1,...$$

$$b_n = \frac{2}{T}\int_{t_0}^{t_0+T} v(t)\sin(n\omega_0 t)dt \qquad n = 0,1,...$$

$$\hat{X} = (A^T A)^{-1} A^T b$$

$$\Gamma_{AA} = \frac{\frac{Z_B}{Z_A} - 1}{\frac{Z_B}{Z_A} + 1} \qquad \Gamma_{BA} = \frac{2\left(\frac{Z_B}{Z_A}\right)}{\frac{Z_B}{Z_A} + 1}$$

$$V(x,s) = E_G(s)\left[\frac{Z_c}{Z_G(s) + Z_c}\right]\left[\frac{e^{-sx/v} + \Gamma_R(s)e^{s[(x/v)-2\tau]}}{1 - \Gamma_R(s)\Gamma_S(s)e^{-2s\tau}}\right]$$

$$I(x,s) = \left[\frac{E_G(s)}{Z_G(s) + Z_c}\right]\left[\frac{e^{-sx/v} - \Gamma_R(s)e^{s[(x/v)-2\tau]}}{1 - \Gamma_R(s)\Gamma_S(s)e^{-2s\tau}}\right]$$

$$\Gamma_R(s) = \frac{\dfrac{Z_R(s)}{Z_c} - 1}{\dfrac{Z_R(s)}{Z_c} + 1} \quad \text{per unit}$$

$$\Gamma_S(s) = \frac{\dfrac{Z_G(s)}{Z_c} - 1}{\dfrac{Z_G(s)}{Z_c} + 1} \quad \text{per unit}$$

$$Z_c = \sqrt{\frac{L}{C}} \ \ \Omega \qquad v = \frac{1}{\sqrt{LC}} \ \ \text{m/s} \qquad \tau = \frac{l}{v} \ \ \text{s}$$